

CMPG 325 COMPUTER NETWORKS

INDIVIDUAL PROJECT

MILESTONE 1: CLIENT REQUIREMENTS AND NETWORK DESIGN

INTRODUCTION

In this project, I am required to analyse design and simulate a computer network for my assigned client, Botshelo Safari Adventures, a tourism organisation based in Rustenburg. The project is part of the CMPG325 Computer Networks individual semester project.

My role is to understand the client's networking requirements and develop a suitable network solution using Cisco Packet Tracer. The network must provide appropriate connectivity and network services while using the assigned IP addressing block 172.30.34.0/23.

The project also includes a specific networking challenge, which is NAT (inside/outside address translation). In addition, I need to consider the requirements for remote management of network devices by an off-site IT contractor and integrate Change Request CR15, which requires a second Internet connection to improve network resilience.

For Milestone 1, I will focus on analysing the client requirements and developing the initial network design. This includes creating the physical and logical topologies, planning the IP addressing scheme and setting up the initial GitHub repository. These design decisions will form the foundation for the Packet Tracer implementation, configuration, testing and the later stages of the project.

CLIENT PROFILE

ITEM	DETAILS
Client ID	CLI-058
Organisation	Botshelo Safari Adventures
Location	Rustenburg
Industry	Tourism
Assigned Address Block	172.30.34.0/23
Technical Challenge	NAT (inside/outside address translation)

Design Constraint	Remote management of network devices is required for an off-site IT contractor
Change Request	CR15: A second Internet connection must be added for resilience and integrated into the network
Implementation	Cisco Packet Tracer

Botshelo Safari Adventures requires a network solution that supports the organisation's internal connectivity requirements and provides a structured, manageable and scalable network environment. As a tourism organisation, the network design must support different areas of business operation while maintaining appropriate separation between network segments and services.

CLIENT REQUIREMENTS

Based on the assigned project brief, the proposed network must meet the following client requirements:

- > Provide appropriate connectivity and network services for the assigned organisation.
- > Use the assigned IP addressing block, 172.30.34.0/23, as the basis for the IP addressing plan.
- > Provide successful communication between the appropriate devices and network segments.
- > Implement the network using Cisco Packet Tracer.
- > Use an appropriate network topology and device arrangement for the assigned scenario.
- > Support logical network segmentation to organise different business areas and services.
- > Allow authorised remote management of network devices for an off-site IT contractor.
- > Provide Internet connectivity for the organisation.
- > Accommodate the assigned technical challenge of NAT using inside and outside address translation.
- > Accommodate Change Request CR15 by integrating a second Internet connection to improve network resilience.
- > Produce a network design that can be implemented, tested, verified and documented throughout the project.

The network design must therefore balance connectivity, segmentation, manageability, resilience and efficient use of the assigned address space.

NETWORK DESIGN REQUIREMENTS

The proposed network will use a structured design consisting of an Internet edge, a central network device for internal connectivity and routing, and access switches for connectivity end devices. The design will separate different business functions and network services into logical network segments.

The network design must support the following functions:

(a) Internal Connectivity

Appropriate internal devices must be able to communicate with authorised network resources. The network will provide connectivity between the relevant departments, services and infrastructure devices.

(b) Network Segmentation

The network will be divided into separate logical segments using VLANs and subnets. This will reduce unnecessary broadcast traffic, improve network organisation and provide a foundation for controlling communication between different areas of the organisation.

Sensitive and specialised areas, such as Finance, network management and guest access, will be separated from general business networks.

(c) Inter-VLAN Connectivity

Where communication between VLANs is required, routing will be provided through the network's Layer 3 infrastructure. This will allow authorised communication between separate logical networks while maintaining network segmentation.

(d) Internet Connectivity

The internal network must connect to an external network through an edge device. This connection will provide the basis for Internet access and for implementing the assigned NAT challenge.

(e) Scalability

The design will use the assigned 172.30.34.0/23 address block efficiently and reserve unused address space for possible future expansion. The network structure will also be designed so that additional devices and services can be incorporated without redesigning the entire network.

(f) Resilience

The design must accommodate a second Internet connection as required by Change Request CR15. The Internet edge will therefore be designed to support both a primary and secondary external connection.

DESIGN CONSTRAINTS

The primary design constraint specified in the project brief is that network devices must be remotely manageable by an off-site IT contractor.

To address this requirement, the proposed network will include a dedicated Network Management segment. Network infrastructure devices such as routers and switches will use management addresses that are logically separated from normal user networks.

This separation is appropriate because network administration should not share the same logical network as general business users or guest devices. The dedicated management segment will provide a controlled foundation for authorised remote administration and troubleshooting.

The exact remote management configuration and security controls will be implemented and verified during the Packet Tracer configuration and testing stages of the project.

TECHNICAL CHALLENGE

The assigned networking challenge for this project is NAT, specifically inside/outside address translation.

The internal network will use private IP addresses from the assigned 172.30.34.0/23 address block. Because these private addresses are not directly routable on an external public network, NAT will be configured at the network edge to translate appropriate internal addresses when communicating with outside networks.

The design will clearly distinguish between:

- > NAT inside interfaces connected to the internal network.
- > NAT outside interfaces connected to external or Internet-facing networks.

The NAT configuration will be implemented, verified and demonstrated in the later implementation phase. Testing will be used to confirm that internal devices can communicate with external networks as required and that address translation is functioning correctly.

CHANGE REQUEST CR15

Change Request CR15 states that a second Internet connection must be added for resilience and integrated into the network.

The proposed network design will therefore include support for two external connections:

- > A primary Internet connection.
- > A secondary Internet connection.

The second connection is intended to improve network resilience by reducing dependence on a single Internet connection. The Internet edge design will be structured so that the additional connection can be integrated without requiring major changes to the internal network topology.

The implementation and verification of the second Internet connection will be addressed during the later stages of the project. However, the requirement is considered during Milestone 1 to ensure that the physical and logical network designs are capable of accommodating the change request.

DESIGN ASSUMPTIONS

The project brief provides the client information, industry, assigned IP address block, technical challenge, design constraint and change request. However, it does not specify the exact number of employees, departments, devices, buildings or network services required by Botshelo Safari Adventures.

For this reason, I made reasonable design assumptions based on the operational needs of a tourism organisation. These assumptions are used to create a realistic and structured network design while ensuring that the network can meet the requirements provided in the project brief.

The following assumptions were made:

- The organisation has different functional areas that require separate network access.
- Reception and Administration staff require access to general internal and external network resources.
- A Reservations and Bookings section is required to manage customer bookings and reservations.
- A Finance section is required and should be logically separated because it may handle sensitive business and financial information.
- Management staff require a separate network segment for management-related activities.
- Safari Operations staff require network connectivity to support the coordination of daily business and safari activities.
- The organisation requires centralised network services.
- Network infrastructure devices require a dedicated management network.
- An off-site IT contractor requires authorised remote access to manage and troubleshoot network devices.
- A guest network is required to provide Internet access while remaining logically separated from the organisation's internal business networks.
- The network should allow for future growth by leaving unused address space within the assigned IP address block.
- The organisation uses a primary Internet connection and must accommodate a secondary Internet connection as required by Change Request CR15.

These assumptions will be used as the basis for the physical topology, logical topology and IP addressing plan.

PROPOSED NETWORK SEGMENTATION

To organise the network effectively, the proposed design separates different business functions and services into individual VLANs. Each VLAN will have its own subnet and default gateway. This creates separate logical network segments and provides a structured approach to network management.

The proposed VLAN structure is shown below:

VLAN ID	VLAN NAME	NETWORK FUNCTION
10	ADMIN	Reception and Administration
20	BOOKINGS	Reservations and Bookings
30	FINANCE	Financial operations
40	MANAGEMENT	Management staff
50	OPERATIONS	Safari and business operations
60	SERVERS	Central servers and network services
70	NET-MGMT	Network infrastructure management
80	GUEST	Guest Internet access

(a) Administration VLAN

The Administration VLAN will support reception and general administrative activities. Devices in this segment will have access to the network resources required for normal business operations.

(b) Bookings VLAN

The Bookings VLAN will support the management of customer reservations and safari bookings. Separating this function from other departments improves network organisation and allows the segment to be managed independently.

(c) Finance VLAN

The Finance VLAN will be used for financial and payment-related activities. It is separated from general business networks because financial operations may involve sensitive information and require more controlled access.

(d) Management VLAN

The Management VLAN will provide a separate network segment for management staff and management-related activities.

(e) Operations VLAN

The Operations VLAN will support staff involved in coordinating safari and daily business operations.

(f) Servers VLAN

The Servers VLAN will contain centralised servers and network services required by the organisation. Separating servers from user networks makes the network easier to manage and provides a structured location for shared services.

(g) Network Management VLAN

The Network Management VLAN will be dedicated to the management of network infrastructure devices such as routers and switches. This VLAN directly supports the requirement for authorised remote management by an off-site IT contractor.

(h) Guest VLAN

The Guest VLAN will provide network and Internet access for guests while keeping guest devices logically separated from the organisation's internal business networks.

JUSTIFICATION OF DESIGN DECISIONS

The proposed design decisions are based on the requirements and constraints specified in the project brief, as well as reasonable assumptions made for a tourism organisation.

(a) Use of VLANs

VLANs are used to separate different departments, services and user groups into individual broadcast domains. This improves network organisation and reduces unnecessary broadcast traffic. VLAN segmentation also provides a foundation for controlling communication between different network segments.

Sensitive areas such as Finance and Network Management are separated from normal business and guest networks. The Guest VLAN is also isolated from internal business networks to reduce the possibility of guest devices accessing internal organisational resources.

(b) Layered Network Structure

The proposed physical topology uses a structured design with an Internet edge, central network infrastructure and access switches for end devices. This approach makes the network easier to organise, manage and expand.

The central network infrastructure will provide connectivity between logical network segments, while access switches will connect end-user devices to the appropriate VLANs.

(c) Dedicated Network Management Segment

A dedicated Network Management VLAN is included because the project requires network devices to be remotely manageable by an off-site IT contractor. Separating network management traffic from normal user traffic provides a more organised and controlled environment for device administration and troubleshooting.

(d) VLSM Addressing

Variable Length Subnet Masking (VLSM) will be used to allocate different subnet sizes according to the expected requirements of the various network segments. This allows the assigned 172.30.34.0/23 address block to be used efficiently and leaves address space available for future growth.

(e) NAT at the Network Edge

The internal network uses private IP addresses from the assigned address block. NAT will therefore be implemented at the network edge to translate internal addresses when devices communicate with external networks. This directly addresses the assigned technical challenge of inside/outside address translation.

(f) Dual Internet Connectivity

The network design accommodates both a primary and secondary Internet connection in response to Change Request CR15. This design improves resilience by reducing the organisation's dependence on a single Internet connection.

The internal network will connect to an edge device that can support both Internet connections. This allows the second connection to be integrated without requiring major changes to the internal VLAN and subnet structure.

(g) Scalability

The network is designed with future expansion in mind. The use of VLANs, a structured physical topology and efficient VLSM addressing makes it possible to add users, devices or additional network services without redesigning the entire network.

PHYSICAL TOPOLOGY

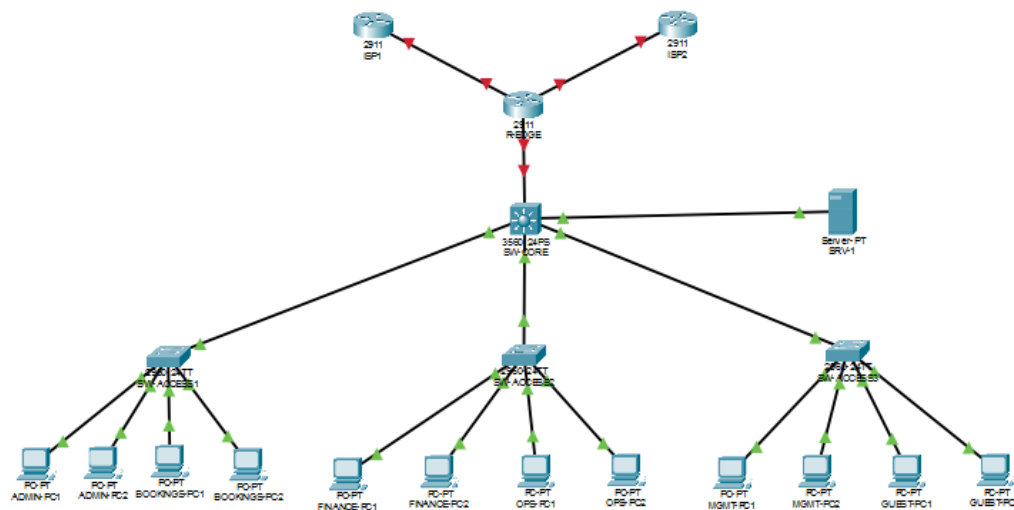


FIGURE1: PROPOSED PHYSICAL TOPOLOGY FOR BOTSHELO SAFARI ADVENTURES

The physical topology shows the network devices used in the proposed network design and how they are physically connected within the organisation. The network follows a hierarchical design consisting of an Internet edge, a core switch and access switches that connect end-user devices.

The topology includes one edge router connected to two Internet Service Providers (ISP1 and ISP2), one multilayer core switch, three access switches, one server and end-user devices representing the different departments within Botshelo Safari Adventures.

The devices are organised as follows:

R-EDGE connects the organisation to the Internet and will later be configured to perform NAT.

ISP1 represents the primary Internet connection.

ISP2 represents the secondary Internet connection required by Change Request CR15.

SW-CORE is the central multilayer switch that connects the internal network infrastructure.

SW-ACCESS1, SW-ACCESS2 and SW-ACCESS3 connect department end devices to the network.

SRV-1 is connected directly to the core switch and represents the organisation's central server.

End-user devices are grouped according to their departments for easier network management.

This physical topology provides a structured network that is easy to manage, supports future expansion and accommodates the project's design constraint and change request.

LOGICAL TOPOLOGY

The logical topology shows how the network is divided into different VLANs and IP subnets. Each department and network function is assigned to its own VLAN to provide logical separation and make the network easier to manage.

The multilayer core switch will provide inter-VLAN routing between the required internal networks. Traffic that needs to reach an external network will be forwarded from the core switch to the edge router, where NAT will be implemented. The edge router connects to both ISP1 and ISP2, with the second Internet connection included to satisfy Change Request CR15.

A dedicated Network Management VLAN is included for the management of network infrastructure devices and to support the remote-management requirement for the off-site IT contractor. Guest devices are placed in a separate Guest VLAN so that guest traffic can be separated from the organisation's internal business networks.

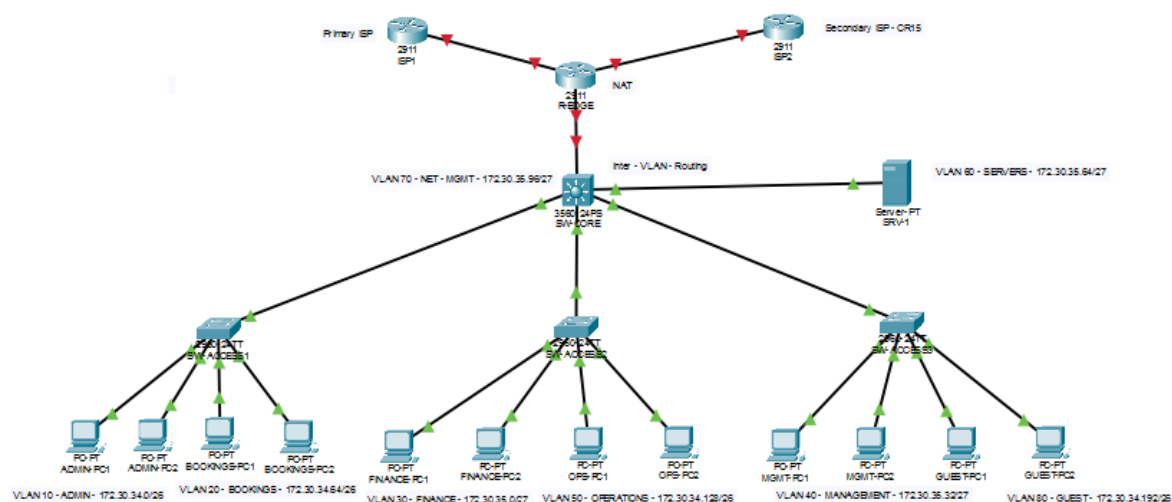


FIGURE2: PROPOSED LOGICAL TOPOLOGY FOR BOTSHELO SAFARI ADVENTURES

IP ADDRESSING PLAN

The assigned IP address block for Botshelo Safari Adventures is **172.30.34.0/23**. I used Variable Length Subnet Masking (VLSM) to divide this address space into separate subnets for the different VLANs. Larger user networks were allocated /26 subnets, while smaller and specialised networks were allocated /27 subnets. The first usable IP address of each subnet is reserved as the default gateway.

VL AN	NAME	NETWOR K ADDRES S	PRE FIX	SUBNET MASK	DEFAULT GATEWA Y	USABLE HOST RANGE	BROADC AST
10	ADMIN	172.30.34.0	/26	255.255.255.192	172.30.34.1	172.30.34.1 - 172.30.34.62	172.30.34.63
20	BOOKING S	172.30.34.64	/26	255.255.255.192	172.30.34.65	172.30.34.65 – 172.30.34.126	172.30.34.127
50	OPERATI ONS	172.30.34.128	/26	255.255.255.192	172.30.34.129	172.30.34.129 – 172.30.34.190	172.30.34.191
80	GUEST	172.30.34.192	/26	255.255.255.192	172.30.34.193	172.30.34.193 – 172.30.34.254	172.30.34.255
30	FINANCE	172.30.35.0	/27	255.255.255.224	172.30.35.1	172.30.35.1 – 172.30.35.30	172.30.35.31
40	MANAGE MENT	172.30.35.32	/27	255.255.255.224	172.30.35.33	172.30.35.33 – 172.30.35.62	172.30.35.63
60	SERVERS	172.30.35.64	/27	255.255.255.224	172.30.35.65	172.30.35.65 – 172.30.35.94	172.30.35.95
70	NET- MGMT	172.30.35.96	/27	255.255.255.224	172.30.35.97	172.30.35.97 –	172.30.35.127

						172.30.3 5.126	
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The addressing plan reserves the first usable address in each subnet for the default gateway. The remaining addresses can be allocated to end devices, servers and other network resources as required. The address range **172.30.35.128 to 172.30.35.255** remains unallocated and is reserved for future network expansion.

Using VLSM allows the assigned /23 address block to be used efficiently because different subnet sizes are allocated according to the expected requirements of each network segment. It also provides sufficient room for future growth without requiring the existing addressing structure to be redesigned.